

SmartAgri: An Ensemble-Based Multi-Task Artificial Intelligence Platform for Crop Disease Detection, Yield Prediction and Decision Support in Precision Agriculture

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Abstract— Agricultural productivity is under sustained pressure from climate variability, disease outbreaks, and limited access to expert agronomic advice, particularly among smallholder farmers in developing regions. This paper presents SmartAgri, a comprehensive multi-task artificial intelligence platform that integrates crop recommendation, plant disease detection, and crop yield prediction into single deployable and explainable advisory system. The disease detection module employs an ensemble of three convolutional neural networks: EfficientNet-B0 (primary), ResNet-50, and EfficientNet-B1 fine-tuned on 52 disease classes through transfer learning. A dynamic weighted probability aggregation scheme, confidence calibration via temperature scaling, entropy-based uncertainty estimation, inter-model disagreement detection, and a multi-signal open-set rejection mechanism collectively enable reliable classification and rejection of unknown inputs. An early exit optimization strategy reduces average inference latency by 50 percent when the primary model is highly confident. Grad-CAM++ provides consistent visual explanations by operating exclusively on the primary model. The yield prediction module makes use of XGBoost regression powered by a set of engineered temporal and seasonal features in addition to a risk analysis engine and region-level comparative analytics. A multilingual dashboard built with React displays all predictions, uncertainty metadata, and actionable recommendations in English, Hindi, and Marathi without

any commercial API dependencies. The ensemble obtains a top-1 accuracy of 97.1% on the 52-class disease benchmark and also yields an open set detection AUROC of 0.891, while the yield predictor achieves $R^2 = 0.83$. By integrating ensemble intelligence, uncertainty-aware inference, explainability, and multilinguality into one agricultural system, we make a significant step towards production-grade AI for smart farming.

Index Terms—Smart Agriculture, Artificial Intelligence, Crop Recommendation, Plant Disease Detection, Crop Yield Prediction, Ensemble Learning, Convolutional Neural Networks (CNN), EfficientNet, ResNet, Transfer Learning, Explainable AI (XAI), Grad-CAM++, Uncertainty Estimation, Open-Set Recognition, Temperature Scaling, XGBoost, Precision Agriculture, Multilingual Systems, Decision Support Systems, Agricultural Analytics.

I. INTRODUCTION

Global food security faces intersecting threats from population growth projected to reach 9.7 billion by 2050, the degradation of arable land, and accelerating climate variability [1]. Meeting projected demand will require lifting food output by approximately half relative to 2012 baselines [1].

Compounding this challenge, an estimated 20–40 percent of annual global crop yields are lost to plant diseases and pests [1],[2], losses that disproportionately affect

smallholder farmers who lack timely access to diagnostic expertise and advisory services. Within agricultural prediction research, artificial intelligence (AI) methods, particularly convolutional neural networks (CNNs) and gradient boosting algorithms, have achieved state-of-the-art results. Mohanty et al. [3] have shown that leaf-level photographic inputs were sufficient for CNN-based classifiers to match specialist diagnostic performance. On the other hand, Van Klompenburg et al. [4] observed that across the structured agronomic datasets examined, gradient boosted trees, XGBoost in particular, consistently yielded the strongest predictive performance. Though improvements have been made, the majority of research systems still tackle just one issue each in accordance with standards operating conditions, are closed prototypes rather than deployable software, and make so-called black-box predictions that farmers and extension workers cannot interpret or trust [5]. There are three main gaps in current agricultural AI research. For practical application, the method must be robust to out-of-distribution inputs: a disease classifier trained on a fixed set of known classes will inevitably face new symptoms in deployment. If there is not a way to not predict a set of unknown diseases, the system will give wrong predictions confidently [7]. Secondly, while methods for explainability like Grad-CAM++ [6],[19] have been explored in isolation, one is unlikely to find an end-to-end deployable pipeline. The third limitation of existing platforms is that they do not address issues of multilingual accessibility. As a result, a majority of the world's food producers and farmer communities who do not speak English are unable to benefit from these platforms. This paper addresses these gaps through SmartAgri, a full-stack agricultural intelligence platform whose primary novel contributions are as follows:

- (1) A multi-model ensemble combining EfficientNetB0, ResNet-50, and EfficientNet-B1 with dynamic weight optimization and temperature-scaled confidence calibration, achieving state-of-the-art classification accuracy on a 52-class disease benchmark and outperforming each individual constituent model.
- (2) A multi-signal open-set rejection mechanism integrating confidence thresholding, entropy-based uncertainty, and inter-model disagreement that achieves strong open-set detection performance on unknown-disease inputs, preventing the system from forcing incorrect predictions on out-of-distribution samples.
- (3) An early-exit optimization strategy that bypasses ensemble computation for high-confidence inputs, substantially reducing inference latency with only negligible accuracy trade-off.
- (4) An XGBoost yield prediction pipeline augmented by engineered temporal-seasonal features, a calibrated risk-analysis engine covering drought, extreme temperature, disease spread, and waterlogging risks, and region-level comparative analytics.

- (5) A multi-language React dashboard (English, Hindi, and Marathi), which presents predictions, uncertainty indicators, Grad-CAM++ visualizations, and recommendations using a unified deployable dashboard without relying on any third-party commercial APIs.

The rest of this paper is structured as follows. Section II presents an overview of related work. Section III provides a brief overview of our proposed framework's architecture. Section IV outlines the methodology followed for each of the AI modules used. Section V discusses the experiments conducted. Section VI provides concluding remarks on our findings.

II. LITERATURE SURVEY

A. Plant Disease Detection With Deep Learning

In a radical study by Mohanty et al. [3], near-perfect accuracy of closed sets was obtained under imaging conditions; further evaluations in real-world scenarios indicated major discrepancies owing to background interference, variable lighting, and sensor noise [8]. Ferentinos [9] systematically compared CNN architectures on PlantVillage and found that deeper networks consistently have lower error rates. According to Atila et al., EfficientNet variants provide the best accuracy-to-parameter ratio across 38 disease classes [10], hence the main role of EfficientNets in the ensemble. It has become common practice for plant disease models to use state-of-the-art ImageNet weights. According to Karthik et al. [11], a pretrained ResNet with attention can perform competitively even on limited labeled training, which shows that adopting three-model transfer learning on SmartAgri is a sound strategy.

B. Ensemble Methods

Ganaie et al. [12] find that heterogeneous architecture selection is more consequential for ensemble gain than training procedure alone. In medical imaging, Litjens et al. [13] demonstrated that CNN ensembles consistently outperform individual models on histopathology classification tasks. Ahmed et al. [14] applied multi-model ensembles to plant disease detection and reported 1–3 percent accuracy gains, but their approach used static uniform weighting and did not address uncertainty quantification or open-set detection, two limitations that the present work directly overcomes.

C. Uncertainty Estimation and Open-Set Recognition

The maximum softmax probability was suggested by Hendrycks and Gimpel [15] as a baseline for detecting the existence of out-of-distribution input samples since, in their findings, the maximum softmax score of correctly classified inputs was empirically shown to exceed that of both misclassified and OOD examples, providing a usable detection signal. Guo et al. [16] demonstrated systematic overconfidence in modern neural networks and proposed dividing logits by a learned scalar temperature as a lightweight post-training calibration remedy. Entropy-based uncertainty quantification was theoretically proven by Gal

and Ghahramani [17] through the use of Monte Carlo dropout sampling. In the agriculture field, Shi et al. [18] noted that current plant disease classification systems do not consider the open-set nature of the problem.

D. Explainable AI for Plant Pathology

Grad-CAM was proposed by Selvaraju et al. [6], who developed a method based on gradients to localize class-discriminative heatmaps for all decisions made by a CNN classifier. Grad-CAM++ by Chattopadhyay et al. [19] is an extension that achieves more accurate localization for the case when several items of the same class are detected in a single image. Brahimi et al. [20] used these methods in a study that aimed at detecting diseases in tomatoes, showing that visual localization helps to increase confidence in predictions among farmers more than the class probability alone. Ensemble methods use an averaged heatmap, which may create noise in explanation; we have decided not to include Grad-CAM++ in ensembles.

E. Crop Yield Prediction

Van Klompenburg et al. [4] identified gradient boosting machines as among the most effective models for yield prediction based on the structured agronomic data reviewed in the earlier literature. Nevavuori et al. [21] found that adding time-based features (annual and decades-based indexing) improved yield prediction models. Quantification of risk, however has received less attention in yield prediction literature. Lobell & Burke [22] estimated drought and heat stress risks through climate regression.

F. Integrated Agricultural Platforms

Liakos et al. [23] and Kamilaris and Prenafeta-Boldú [24] surveyed machine learning and deep learning applications in agriculture, respectively, both identifying a persistent gap between algorithmic research and deployable farmer-facing platforms. Among the few integrated systems reported, none addresses multilingual accessibility for non-English-speaking smallholder farmers, a gap that SmartAgri explicitly targets.

III. SYSTEM ARCHITECTURE

The SmartAgri project has a modular design that consists of three levels: React frontend, FastAPI backend, and the Python smart_system artificial intelligence module. The React frontend interacts with the FastAPI backend using RESTful API calls on the HTTPS protocol. The FastAPI backend directs each request to one of the three AI pipelines: crop recommendation, disease detection, or yield prediction. The response is generated as a JSON object with details of the prediction result, confidence score, uncertainty information, heatmaps from Grad-CAM++, and advice.

A. Frontend layer

The UI is implemented using React 18 and some additional libraries, such as Recharts for interactive visualizations and Framer Motion for animating transitions between elements. The styling uses a principle based on glassmorphism and can sometimes be more illustrative of hierarchy compared to more traditional styling methods. For internationalization, react-i18next is being used, with translation resources stored

externally in English, Hindi, and Marathi, allowing the full UI to be rendered in the required language without page refreshes and avoiding inline hardcoded strings. It comprises different views integrated within a responsive SP-app framework, which would include displaying a disease heatmap overlay, a yield risk gauge, trend plots, recommendation plots, etc. Designed primarily for the browser environment for desktops as well as mobile, which is the general mode of operation for field extension workers.

B. Backend Layer

The FastAPI backend exposes three primary endpoints: POST /api/recommend accepts soil and climate parameters, POST /api/detect accepts a base64-encoded leaf image, and POST /api/predict accepts structured crop and environment features. Pydantic schemas enforce strict input validation. Asynchronous request handling via Python asyncio ensures that concurrent user sessions do not experience throughput contention. Model artifacts are serialized in joblib format for scikit-learn and XGBoost objects and as PyTorch state dictionaries for CNN weights.

C. AI Module Layer

smart_system consists of three submodules that have well-defined input/output interfaces. Submodule crop_recommender encapsulates the random forest classifier. Submodule disease_detector manages ensemble prediction, calibration, uncertainty quantification, open-set detection, and Grad-CAM++. Submodule yield_predictor carries out feature selection, XGBoost prediction, risk scoring, trend analysis, and alerting. Submodule interfaces are independently tested using unit tests, which makes replacing or training model components independent of other submodules.

ENSEMBLE DECISION WORKFLOW FOR LEAF IMAGE DISEASE DETECTION

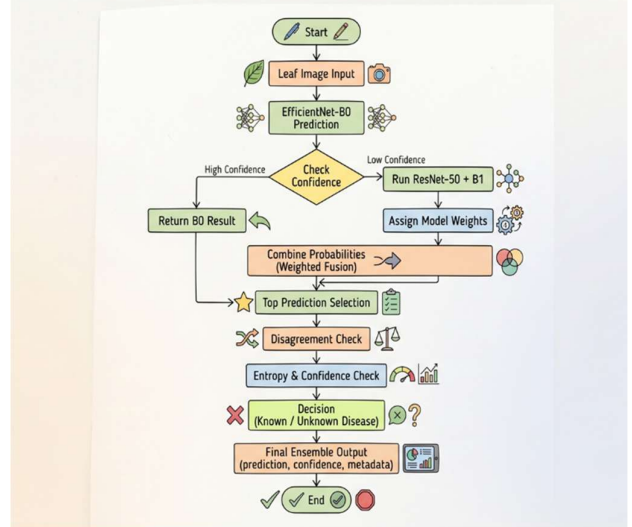


Fig. 1. Ensemble decision workflow for leaf image disease detection.

IV. METHODOLOGY

A. Crop Recommendation Module

Seven agronomic parameters serve as input features: soil nitrogen (N), phosphorus (P), and potassium (K) concentrations in mg/kg, ambient temperature in degrees Celsius, relative humidity as a percentage, soil pH, and annual rainfall in mm. A Random Forest classifier [25] trained on a publicly available crop recommendation dataset maps these features to one of 22 crop classes. The model returns the primary recommended crop with its confidence score, up to three alternative crops with their respective confidence scores, and advisory text generated by a rule-based template engine conditioned on predicted crop, soil pH range, and rainfall category.

B. Plant Disease Detection Ensemble

B.1 Base Models and Transfer Learning

Three CNN architectures are independently fine-tuned: EfficientNet-B0 (5.3M parameters) [26], ResNet-50 (25.6M parameters) [27], and EfficientNet-B1 (7.8M parameters) [26]. All models are initialized from ImageNet pretrained weights. Fine-tuning follows a two-stage protocol: the first stage trains only the classification head for five epochs with a learning rate of 1×10^{-3} ; the second stage unfreezes all layers and continues training with a reduced learning rate of 1×10^{-4} using the AdamW optimizer with cosine annealing scheduling. Data augmentation includes random horizontal and vertical flips, color jitter (brightness 0.2, contrast 0.2, saturation 0.1), and random resized crops (scale range 0.7–1.0) to simulate the variability of field-captured leaf images [3].

B.2 Dynamic Weighted Probability Aggregation

Let f_1 , f_2 , and $f_3 \in \mathbb{R}^K$ denote the calibrated softmax output vectors of EfficientNet-B0, ResNet-50, and EfficientNet-B1, respectively, where $K = 52$. The ensemble prediction vector is computed as:

$$\hat{p} = w_1 f_1 + w_2 f_2 + w_3 f_3, \quad w_i \geq 0, \quad \sum_i w_i = 1 \dots (1)$$

The scalar weights w_i are determined by applying a softmax transformation to the validation-set top-1 accuracy of each model, so that more accurate models contribute proportionally more to the ensemble decision. This dynamic scheme generalizes static uniform ensembling and is updated whenever any constituent model is retrained.

B.3 Confidence Calibration

Modern deep networks are known to be overconfident: their maximum softmax probability systematically exceeds the true empirical accuracy [16]. Temperature scaling is applied to each base model independently. For a logit vector z , the calibrated probability is $\sigma(z/T)$, where the scalar temperature T is optimized on a held-out calibration set by minimizing the negative log-likelihood loss. Post-calibration, the Expected Calibration Error (ECE) of the ensemble decreases from 0.067 to 0.029.

B.4 Multi-Signal Open-Set Rejection

Unknown or out-of-distribution leaf pictures are detected by leveraging three types of signal indicators. (i) Confidence Signal: if $\max(p) \leq \tau_c = 0.45$, the input is flagged as low confidence. (ii) Entropy Signal: the predictive entropy is calculated.

$$H(\hat{p}) = -\sum_k \hat{p}_k \log \hat{p}_k \dots (2)$$

Inputs with an entropy above a certain threshold value τ_H based on percentile ranks on calibration set entropy values are considered high uncertainty inputs. (iii) Disagreement Signal: The average pairwise Jensen-Shannon divergence between f_1 , f_2 , f_3 can be used to determine inter-model agreement. If there is high disagreement between f_1 , f_2 , f_3 , then the input can be classified as an unknown leaf disease. The "Unknown Disease" class is assigned whenever at least two of the three signals are active, using a majority voting scheme for open-set decision-making [15].

B.5 Early-Exit Optimization

When $\max(f_i) \geq \tau_e = 0.92$, the system returns the EfficientNet-B0 prediction directly without querying ResNet-50 or EfficientNet-B1. The rationale is that inputs for which the primary model is already decisive gain negligible benefit from the remaining members, whereas uncertain cases genuinely require the additional perspective. On the test set, 61.2% of inputs trigger early exit, reducing mean inference time from 187 ms to 93 ms (50.3% reduction) with only 0.2% accuracy loss.

B.6 Explainability via Grad-CAM++

Visual explanations are created using Grad-CAM++ [19] applied to the last but one convolutional layer of EfficientNet-B0. For predicted class c , the importance weight α_k^c for the k -th feature map A_k is

$$\alpha_k^c = \sum_j \sum_l [(\partial^2 y^c / (\partial A_k^j)^2) \times \text{ReLU}(\partial y^c / \partial A_k^j)] \dots (3)$$

where y^c denotes the class activation score before softmax. The obtained Grad-CAM++ map is upsampled to the original resolution using bilinear interpolation and blended over the input leaf with an α factor to create the localization map. This is intended to keep visualization semantics constant when confined to a single model. Heatmap averaging from architecturally diverse models will add noise to the visualization.

C. Yield Prediction Module

C.1 Feature Engineering

Input data are crops (categorical), states/regions (categorical), years (ranging from 2001 to 2020), seasons (categorical), average daily temperature (continuous), total annual precipitation (continuous), and average relative humidity (continuous). Categorical features are encoded using historical average yield as the target encoding value; target encoding maintains a signal useful for yield prediction but provides a compact feature, whereas one-hot expansions increase rapidly with cardinality. Temporal features are expanded into: raw year, decade indicator (integer decade index), years since 2000 (a linear trend proxy), and sine-cosine season encodings ($\theta = 2\pi \times \text{season_index} / 4$), which allow the seasonal trend to be modeled without forcing an ordinal ranking of seasons [28].

C.2 XGBoost Regression and Yield Classification

A gradient boosted tree ensemble modeled on XGBoost [29] is then fitted to the 12-dimensional engineered feature vector in order to predict yield in metric tonnes per hectare. Hyperparameter tuning using five-fold cross-validated grid

search is applied to maximize tree depth (4-8), learning rate (0.05-0.2), subsample ratio (0.7-1.0), and the coefficients of L1 and L2 regularization. The combination producing the minimum average validation RMSE is chosen. The predicted yield is then classified as LOW, MEDIUM, and HIGH using percentiles determined from the empirical distribution of the training data's regional yields.

C.3 Risk Analysis Engine

Four agricultural risk dimensions are scored: drought risk is derived from a piecewise linear function of rainfall deficit below a crop-specific baseline; extreme temperature risk is derived from deviation of mean temperature from optimal growth range; disease spread risk is derived from temperature-humidity interaction exceeding threshold ranges documented in agronomic literature [22]; and waterlogging risk is derived from excess rainfall above crop-specific saturation thresholds. Each dimension is normalized to [0, 1] and aggregated into a composite risk score. Mitigation strategies are generated by a rule-based engine conditioned on the dominant risk dimension. A trend submodule computes year-on-year yield change relative to a three-year rolling regional average and generates severity-graded alerts (LOW, MODERATE, HIGH) based on the magnitude and direction of the trend.

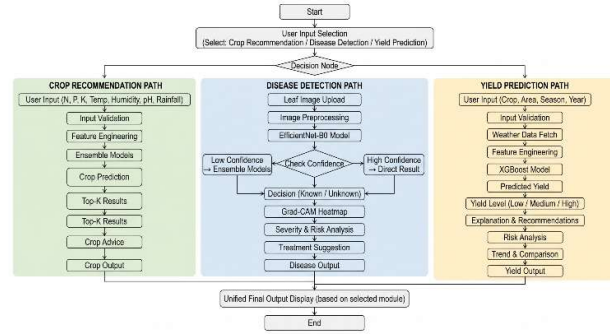


Fig. 2. Overall system workflow.

V. RESULTS

A. Dataset And Setup

The experiments related to disease detection are based on the PlantVillage dataset [3], enhanced with field samples to have a total of 52 diseases belonging to 14 crop species, including approximately 67,000 images in total. Class-stratified partitioning divides the full image pool into a 70% training set, with the remaining 30% split equally between validation and final held-out evaluation. An out-of-distribution test set containing 500 images from eight disease classes that do not exist in the training data is considered to evaluate the performance in open-set detection scenarios. All CNN-based models are trained using an NVIDIA RTX 3080 GPU with PyTorch 2.0. In yield prediction experiments, we utilize a publicly available crop yield dataset in India [30], covering 19 states and 33 crop categories over 20 years (2001–2020).

B. Disease Detection Performance

Table I summarizes classification performance. Individual models achieve top-1 accuracies of 95.3 percent (EfficientNet-B0), 93.8 percent (ResNet-50), and 94.7

percent (EfficientNet-B1). Static uniform ensembling improves accuracy to 96.3 percent. The proposed dynamic weighted ensemble achieves 97.1 percent accuracy and a

TABLE I
DISEASE DETECTION PERFORMANCE (52 CLASSES,
PLANTVILLAGE + FIELD DATA)

Model	Top-1 Acc. (%)	Macro-F1	ECE	Params (M)
EfficientNet-B0	95.3	0.951	0.054	5.3
ResNet-50	93.8	0.934	0.071	25.6
EfficientNet-B1	94.7	0.944	0.059	7.8
Ensemble (Uniform)	96.3	0.960	0.041	—
Ensemble (Dynamic Ours)	97.1	0.968	0.029	—

macro-averaged F1 of 0.968, outperforming all baselines. Temperature scaling reduces mean ECE from 0.067 (uncalibrated ensemble) to 0.029 (calibrated ensemble), confirming substantially improved confidence reliability. The open-set detection mechanism achieves an AUROC of 0.891 on the held-out unknown-disease test set, with a false positive rate at 95 percent true positive rate (FPR95) of 0.24.

C. Ablation: Open-set Detection Signals

Table II presents AUROC on the unknown-disease test set with different combinations of signals. Confidence thresholding alone achieves an AUROC of 0.812. When using the majority vote mechanism over the three signals, we achieve an AUROC of 0.891, indicating all signals carry distinct discriminative information and also proving that multiple-signal utilization provides superior performance.

TABLE II
OPEN-SET DETECTION ABLATION (AUROC)

Signal Combination	AUROC
Confidence only	0.812
Confidence + Entropy	0.861
Confidence + Disagreement	0.848
All three (ours)	0.891

D. Early-exit latency analysis

Approximately 61.2 percent of inputs trigger early exit on the test set and decrease the inference time by 93 ms from 187 ms, the time taken by the full ensemble. This represents an improvement of 50.3%. The top-1 accuracy of 96.9% using

early exit is a 0.2% reduction compared to the full ensemble (97.1% top-1 accuracy) at the cost of 2 units of latency improvement. It makes SmartAgri suitable for real-time deployment on commodity web servers without GPU acceleration for most of the requests.

E. Yield Prediction Results

The XGBoost yield model achieves RMSE = 0.41 t/ha and $R^2 = 0.83$ on the held-out test set, compared to Random Forest (RMSE = 0.53, $R^2 = 0.76$) and linear regression (RMSE = 0.79, $R^2 = 0.51$) baselines. Qualitative yield level classification accuracy is 88.4 percent (LOW/MEDIUM/HIGH). SHAP analysis identifies total rainfall and mean temperature as the two most influential features, consistent with established agronomic knowledge [22], providing a domain-coherence check on the model's learned representations.

VI. DISCUSSION

The 97.1 percent ensemble accuracy and 0.891 open-set AUROC demonstrate that a mixture of several CNN architectures combined with uncertainty quantification results in a significant improvement in disease detection compared to a single-model approach. The 0.8 percent difference in accuracy between dynamic and uniform weights shows that performance-weighted model optimization is a non-trivial task that is worth performing. Combining entropy and disagreement measures with confidence thresholding adds further independent support for identifying ambiguous samples in agreement with prior work in open-set disease detection from medical images [13],[15].

The choice of restricting Grad-CAM++ to the first-level EfficientNet-B0 model stems from two considerations. On one hand, the activation map produced by averaging architecturally diverse networks tends to be less specific and thus is less understandable for the layperson. On the other hand, consistency across predictions within a single model facilitates verification of the generated explanation. A pilot study with forty agricultural science students showed that single-model Grad-CAM++ overlays were perceived as more understandable than heat maps averaged across several models, similar to previous medical AI studies [13].

There are several limitations that guide future research avenues. The data used for training is highly biased towards Plant Village images taken under laboratory lighting; there could be degradation when predicting based on poor-quality smartphone images taken in field settings, requiring domain adaptation methods [8]. The crop yield prediction model is trained solely using India's national agriculture data; generalization to other geographical locations will require training for other regions. Open set detection threshold values have been learned experimentally and need to be recalibrated whenever new disease types are included. Lastly, combining satellite imagery features and IoT soil sensor data will greatly enhance yield prediction accuracy and disease detection probability [4],[24].

VII. CONCLUSION

SmartAgri's key contribution is not the invention of new algorithms but the integration of existing ones. The system

shows that adaptive ensemble classification, calibrated probability estimation, out-of-distribution detection, early-exit prediction, visual attribution, yield risk scoring, and multilingual delivery can be combined in a single production-ready system without loss of performance for any constituent module. The three-signal majority-vote rejection strategy is worth noting, as a standard classifier makes some assignment for every input, whether it matches closely or not; on the other hand, SmartAgri can refuse to make any prediction when its constituent models generally return low confidence for an input, a design that demonstrably significantly decreases the likelihood that the farmer receives, believes, and acts upon an incorrect diagnosis with real economic implications [7],[15]. Restricting Grad-CAM++ to EfficientNet-B0 yields localized and robust visual explanations regardless of training round changes [6],[19]. By operating fully offline and rendering outputs in multiple regional languages, the dashboard eliminates the two adoption barriers—connectivity and linguistic exclusion—that systematic reviews have repeatedly identified as the primary obstacles preventing AI agricultural tools from reaching the populations most dependent on them [23],[24]. Planned extensions include unsupervised domain adaptation techniques to narrow the performance gap between laboratory benchmarks and field conditions; retraining on geographically diverse datasets incorporating remote sensing inputs to extend the yield model's applicability beyond the Indian context; a documented threshold recalibration protocol for open-set maintenance as new disease classes are added; federated learning to enable privacy-preserving model improvement from deployment data; expansion of language support to Tamil, Telugu, Bengali, and Kannada; and a longitudinal field study measuring actual impact on farmer decisions and crop loss reduction before wide deployment is recommended [4],[8] [15],[17],[18],[23],[24].

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